

HASSAN RAZA

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PROFESSIONAL SUMMARY

Gold Medal-winning Electrical Engineering Graduate (Computer Systems) with a strong foundation in Digital Design, RTL Development, Computer Architecture, and ASIC Design Flow. Experienced in developing single-cycle and pipelined RISC-V processors using Verilog, performing simulation and waveform analysis with GTKWave, and gaining hands-on exposure to the OpenROAD physical design flow, including Floorplanning, I/O Placement, PDN, and Static Timing Analysis (STA). Proficient in Ubuntu Linux, Git/GitHub, and VS Code, with internship experience in semiconductor design and verification workflows. Seeking an RTL Design Engineer or Design Verification Engineer role to contribute to next-generation digital IC and SoC development.

EDUCATION

UIT University — BE Electrical Engineering (Computer Systems) July 2026 Karachi, Pakistan

CGPA: 3.44

Gold Medalist

WORK & PRACTICAL EXPERIENCE

IC Design Trainee | Chip Slect (PSEB) May, 2026 - Ongoing

- Developed digital hardware modules including UART, VGA, AHB, APB, AXI-LITE using Verilog.
- Implemented and verified digital designs on ARTY-A7 and ZYBO FPGA development boards.
- Simulated and analyzed digital hardware designs using **Verilog**, **QuestaSim**, and **GTKWave**.
- Used **OpenROAD** to explore **Floorplanning**, **I/O Placement**, **PDN**, **Static Timing Analysis (STA)** reports, and configure design flows through config.yaml.

IoT Intern | NACI Lab NED University 2026

- Contributed to the development of the OBDCharm IoT Vehicle Monitoring System.
- Worked with embedded hardware, ESP32, sensors, and real-time data acquisition.
- Assisted in IoT communication and embedded software integration.
- Participated in testing and validation of embedded applications.

Electrical Engineering Intern | Pakistan International Airlines (PIA) 2024

- Observed aircraft electrical maintenance procedures and technical documentation.
- Assisted engineers in preventive maintenance and inspection activities.
- Gained practical understanding of aviation electrical systems and maintenance workflows.

Research Intern | Merl 2023

- Designed and simulated digital hardware modules using Verilog and GTKWave.
- Developed **single-cycle and pipelined RISC-V processors** using **Verilog** in **Visual Studio Code**.
- Verified processor functionality through **simulation and waveform analysis** using **GTKWave** in a **Ubuntu Linux** environment.

WORK & PRACTICAL EXPERIENCE

President | UITU Maker's Club 2024-2026

- Led technical workshops, engineering events, and student innovation activities.
- Coordinated project development and encouraged collaborative learning among engineering students.
- Managed club operations while promoting embedded systems and hardware design initiatives.

PROJECTS

Smart Swarm Power Distribution & Monitoring System

ESP32, IoT, Arduino IDE, Wi-Fi Designed an IoT-enabled smart energy distribution system capable of real-time monitoring, wireless communication, and scalable energy management using ESP32 microcontrollers.

Uart Controller Design

- Designed and implemented a UART (Universal Asynchronous Receiver/Transmitter) controller using Verilog.
- Verified serial data transmission and protocol functionality through waveform analysis using GTKWave.

VGA Controller with SRAM

- Developed a **VGA controller interfaced with SRAM** on an **FPGA** to display graphical output on a monitor.
- Implemented memory read operations and verified timing and display functionality using **GTKWave**.

ASIC Physical Design Flow using OpenROAD

- Performed the ASIC physical design flow for the **SPM (Single-Port Memory)** design using **OpenROAD** and **Yosys**.
- Analyzed **RTL synthesis**, **floorplanning**, **I/O placement**, **STA**, **area reports**, **metrics.csv**, **pin configuration**, and other implementation reports to evaluate the physical design.

TECHNICAL SKILLS

Programming Languages: Python, C, C++, Python, Verilog , systemverilog

Embedded System & Hardware: ESP32, Arduino, Raspberry Pi, FPGA (ARTY-A7, ZYBO), RISV-V Architecture, Digital Logic Design

sFPGA & EDA Tools: Questasim,OpenRoad,GTKWave,

Software & Tools: Ubuntu Linux, MATLAB , Proteus , AutoCAD , CiscoPacketTracer , AWS , OpenStack

Database: MySQL

CERTIFICATIONS & COURSES

- Hardware Design Training – CHIP SELECT (PSEB)
- IoT Internship – NCAI Lab, NED University
- PITP Cloud Computing Program
- Digital Freelancing Course